VentureInsight SQL Investment Analysis

Project Overview

This project focuses on analyzing venture capital and startup investment data using SQL. The analysis was performed as part of a simulated business case for **VentureInsight**, a research firm that provides insights to venture capital firms and startup investors.

The objective was to answer real-world business questions related to startup funding, acquisitions, investor behavior, fund activity, and employee education trends using SQL queries and relational database analysis.

Through this project, I worked with multiple interconnected tables and applied SQL techniques such as filtering, aggregation, grouping, joins, subqueries, and conditional logic to generate meaningful business insights.

Table	Description
company	Startup information including funding, status, and category
fund	Venture capital fund details
funding_round	Information about funding rounds
investment	Investment transactions between funds and startups
acquisition	Company acquisition records
people	Employee, founder, and investor information
education	Educational background of individuals

Business Questions

The analysis focused on answering the following questions:

1. How many startups in the dataset have failed or closed?
2. Which U.S. news companies raised the most funding?
3. How much money was spent on cash acquisitions between 2011–2013?
4. Which industry influencers have strong Twitter branding?
5. Which finance-related influencers match specific criteria?
6. Which countries attracted the most startup funding?
7. Were there large fluctuations in funding activity on certain dates?
8. How active are venture funds based on investment volume?
9. Do highly active funds participate in more funding rounds?
10. Is there a relationship between employee education and startup failure?

Tools & Skills Used

- SQL
- PostgreSQL
- Data Aggregation
- Filtering & Sorting
- GROUP BY & HAVING
- CASE Statements
- Subqueries
- INNER JOIN
- Business Analysis
- Relational Database Analysis

SQL Analysis & Queries

1. Startup Landscape Analysis

Objective

Determine how many companies in the dataset have closed down.

```
SELECT COUNT(*) AS closed_companies   FROM company  WHERE status = 'closed';
```

Insight

This analysis helped establish a baseline understanding of startup failure rates within the dataset.

2. Sector Analysis for U.S. Investors

Objective

Identify funding amounts raised by U.S.-based news companies.

```
SELECT name,  funding_total

FROM company

WHERE country_code = 'USA'

AND category_code = 'news ORDER BY funding_total DESC;
```

Insight

The results highlighted which media and news startups attracted the highest investment funding.

3. Cash Acquisition Analysis

Objective

Calculate the total amount spent on cash acquisitions between 2011 and 2013.

```
SELECT SUM(price_amount) AS total_cash_acquisitions

FROM acquisition

WHERE term_code = 'cash'

AND EXTRACT(YEAR FROM acquired_at) BETWEEN 2011 AND 2013;
```

Insight

This analysis provided visibility into acquisition trends during the post-recession recovery period.

4. Identifying Industry Influencers

Objective

Find individuals whose Twitter usernames start with “Silver”.

```
SELECT first_name,

    last_name,

    twitter_username

FROM people

WHERE twitter_username LIKE 'Silver%';
```

Insight

This helped identify industry personalities with strong social media branding.

5. Finance Influencer Analysis

Objective

Find people whose Twitter usernames contain “money” and whose last names start with “K”.

```
SELECT *

FROM people

WHERE twitter_username LIKE '%money%'
```

```
AND last_name LIKE 'K%';
```

Insight

This query narrowed down finance-focused influencers for targeted outreach analysis.

6. Geographic Investment Analysis

Objective

Calculate total startup funding by country.

```
SELECT country_code,  SUM(funding_total) AS total_funding

FROM company

GROUP BY country_code

ORDER BY total_funding DESC;
```

Insight

The results showed which countries attracted the highest levels of venture capital investment.

7. Funding Round Volatility Analysis

Objective

Compare the highest and lowest funding amounts raised on each date.

```
SELECT funded_at,

    MAX(raised_amount) AS highest_funding,  MIN(raised_amount) AS lowest_funding

FROM funding_round

GROUP BY funded_at

HAVING MIN(raised_amount) <> 0

AND MIN(raised_amount) <> MAX(raised_amount);
```

Insight

This analysis identified dates with significant variation in startup funding activity.

8. Fund Activity Classification

Objective

Classify venture funds based on investment activity.

```
SELECT *,
CASE
  WHEN invested_companies >= 100 THEN 'high_activity'
  WHEN invested_companies >= 20 THEN 'middle_activity'
  ELSE 'low_activity'
END AS activity_level
FROM fund;
```

Insight

The classification helped categorize funds by investment scale and market activity.

9. Investment Strategy by Fund Activity

Objective

Calculate the average number of funding rounds participated in by each activity category.

```
SELECT activity_level,
    ROUND(AVG(funding_rounds)) AS avg_funding_rounds
FROM
(
  SELECT f.id,
      CASE
        WHEN f.invested_companies >= 100 THEN 'high_activity'
        WHEN f.invested_companies >= 20 THEN 'middle_activity'
        ELSE 'low_activity'
      END AS activity_level,
      COUNT(i.funding_round_id) AS funding_rounds
```

```
  FROM fund AS f

  JOIN investment AS i

  ON f.id = i.fund_id

  GROUP BY f.id

) AS activity_data

GROUP BY activity_level

ORDER BY avg_funding_rounds;
```

Insight

The analysis showed how investment behavior differs between highly active and less active venture funds.

10. Employee Education & Startup Success Analysis

Objective

Analyze educational backgrounds of employees working at failed startups that closed after only one funding round.

```
SELECT AVG(t.total_degree_type)

FROM

(

  SELECT p.id,

         COUNT(e.degree_type) AS total_degree_type

  FROM people AS p

  JOIN education AS e

  ON p.id = e.person_id

  WHERE company_id IN

  (

    SELECT id

    FROM company

    WHERE id IN
```

```
    (
      SELECT company_id
      FROM funding_round
      WHERE is_first_round = 1
      AND is_last_round = 1
    )
    AND status = 'closed'
  )
  GROUP BY p.id
) AS t;
```

Insight

This analysis explored whether employee education levels showed any correlation with startup failure.

Key Learnings

1. Through this project, I strengthened my understanding of:
2. Writing complex SQL queries
3. Multi-table joins and relationships
4. Business-driven data analysis
5. Aggregation and grouping techniques
6. Subqueries and nested filtering
7. Conditional logic using CASE statements
8. Converting business questions into analytical solutions